FAISAL SHARIFF

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EDUCATION

San Jose State University Expected May 2028

B.S., Computer Science and Linguistics | San Jose, CA

EXPERIENCE

Research Assistant, RAG-based Conversational AI Mar 2026 – May 2026

San Jose State University | San Jose, CA

- Fine-tuned BERT-based NLU models on a rebuilt pipeline, raising intent accuracy to 92.5% and entity accuracy to 84.5%.
- Built a multimodal RAG chatbot (T5 + Neo4j knowledge graph + TF-IDF retrieval), reaching a 0.60 BLEU score.

SJSU Robotics Club, Intelligence Systems Subteam Aug 2026 – Present

San Jose State University | San Jose, CA

- Design and maintain a Jetson-based sensor test bench fusing 4 sensing modalities (depth camera, lidar, RGB camera, IR) to validate perception reliability, including in low-visibility conditions, ahead of deployment on the club's main robot, Alicanto.
- Built a tightly-coupled LiDAR-inertial localization stack fusing 3 sensor streams (Unitree L2 4D LiDAR, RealSense RGB-D, 250 Hz IMU) by integrating Point-LIO odometry into RTAB-Map's TF tree for loop closure and mapping.
- Eliminated a tilted, smeared 3D SLAM map on a ROS2 Humble autonomous rover, restoring level, coherent reconstruction, by re-architecting the RTAB-Map pipeline to use IMU-fused Point-LIO odometry instead of drift-prone LiDAR-only ICP.
- Containerized the full perception/SLAM pipeline for Apple Silicon (arm64 Docker) into a one-command sim harness that replays real rosbag data into Foxglove, collapsing a 6-step manual bring-up into a single command.

Videographer & Video Editor Jun 2026 – Aug 2026

Step N Groove | Sharjah, UAE

- Edited short-form social and promotional video content in Adobe Premiere Pro on a Monday-Friday schedule, averaging 10-15 finished clips per day.
- Produced promo reels averaging roughly 5,000 views each, working directly from raw studio footage to final cut.

Mental Health Awareness Intern May 2024 – Jun 2024

Delhi Private School Sharjah | Sharjah, UAE

- Created digital posters to raise awareness about mental health support within the school community.
- Helped file and organize administrative documents, and assisted in recording and editing videos for significant school events.

PROJECTS

ATMO: Drone Atmosphere Profiler Oct 2026

Python, FastAPI, I2C, Raspberry Pi, Grove SPA06-003 barometer | GitHub

- Gave a drone live in-flight air data (altitude, climb rate, turbulence) by building a Raspberry Pi barometer payload that sampled pressure at 5 Hz, filtered out noisy spikes with a median-of-5 + EMA filter, and refreshed a FastAPI dashboard every 1 s over its own offline wifi hotspot, so no internet was needed (Cal Hacks 13.0).

Syllabi: AI Deadline Extraction Tool Jul 2026 – Aug 2026

Python, FastAPI, Gemini API, Google Cloud Run, Netlify | GitHub

- Built to automate deadline extraction from course syllabi into calendar reminders after manually tracking due dates across classes proved error-prone, deploying a Python/FastAPI/Gemini API pipeline to Google Cloud Run that generates .ics files, adopted by 10+ students.
- Built a dedicated evaluation harness to check extractions against ground-truth deadlines, which caught a real model hallucination bug and drove the fix, strict JSON schema validation plus prompt constraints, that raised recall to 89%.

Drone Control Pipeline: Raspberry Pi Zero 2W + Betaflight FC Nov 2025 – Feb 2026

Python, UART/USB, Betaflight, Raspberry Pi | GitHub

- Built after the drone's stock RC receiver failed, engineering a Raspberry Pi Zero 2W bridge to translate wireless gamepad input into Betaflight flight-controller commands over UART/USB and restore full flight control.
- Measured end-to-end input latency and drove it down to 20-50ms by switching from TCP to UDP to eliminate head-of-line blocking and adding threading.Lock to prevent race conditions during serial communication.

DriftWatch: Semantic Drift Detection System Aug 2026 – Sep 2026

Python, FastAPI, PostgreSQL/pgvector, sentence-transformers, Chart.js, RQ/Redis | GitHub

- Built for the DevNetwork API World Hackathon to automatically flag when a public figure's or company's news coverage shifts in tone or topic, a shift that's slow and unreliable to catch by manual monitoring.
- Measured drift as cosine distance from a per-entity sentence-transformer embedding centroid stored in pgvector, validating the scoring approach across 370+ real-world articles pulled live via SerpApi and surfaced through a FastAPI backend and Chart.js dashboard.

Autonomous Drone Companion In Progress

C++17, Python, ArduCopter 4.7.2, MAVLink2, mavlink-router, systemd, ArduPilot SITL, Raspberry Pi | GitHub

- Two-person autonomous drone build: 5-inch, 3S quadcopter (MicoAir405v2, ArduCopter 4.7.2) with a Raspberry Pi Zero 2W companion computer streaming live MAVLink data to QGroundControl and C++17 telemetry software, tested in ArduPilot SITL.
- Verified radio and GCS failsafes trigger in about 2 seconds on a joystick or Wi-Fi drop and clear when the link returns , after tracing a missing I2C compass (QMC5883P) to Copter 4.5.7/4.6.0 builds that exclude it on 1MB boards and upgrading to 4.7.2.
- Built the Pi Zero 2W companion that relays MAVLink between the flight controller (USB, 921600 baud) and QGroundControl over Wi-Fi for Xbox-controller flight, compiling mavlink-router from source as an auto-restarting systemd service that survives reboots, fixing a USB hang with the dwc2 host overlay over the legacy dwc_otg driver, and writing dependency-free Python tools that hand-build MAVLink2 packets to read sensor health and parameters.

Rescue Rover: Stuck RC Car Rescue Robot Feb 2026 – Nov 2026

Python, Raspberry Pi, PCA9685, YOLOv8, Roboflow

- Built a rover that rescues stuck RC cars, reaching 95.8% mAP@50-95 on a held-out test split by training YOLOv8 on a 284-image dataset (self-collected plus Roboflow Universe) to detect stuck cars, then deploying it on a Raspberry Pi that drives PCA9685-controlled steering/throttle servos from live camera feedback in real-time rescue test runs.

MenuMind: Graph + RAG Restaurant Chatbot Hack Hayward 2026

Neo4j, Perplexity Sonar API

- Built a Flask + Neo4j backend that parses an uploaded menu (PDF/image) into structured dish nodes (name, price, category, allergens) via Gemini, then answers customer questions with Perplexity Sonar constrained by a system prompt to reference only dishes present in that graph, blocking invented dishes/prices/allergens.
- Added per-route rate limiting (Flask-Limiter: 5 uploads/min, 200 req/day) to protect the Gemini/Perplexity API budget, shipped in a 19-hour hackathon build.

AccessCanvas: AI Accessibility Auditor Apr 2026

React, Flask, Docker, Tesseract OCR, OpenCV, Gemini API | GitHub

- Ran OCR across three preprocessed variants of each image (contrast-boosted, inverted, mean-luminance thresholded) to catch text a single Tesseract pass misses, then deduplicated overlapping detections by bounding-box overlap before scoring.
- Implemented the WCAG 2.2 relative-luminance and contrast-ratio formulas directly against sampled foreground/background pixels to flag text under the 4.5:1 (normal) / 3:1 (large-text) thresholds, backed by a Gemini alt-text generator with a mechanical fallback if the API call fails.

Sleep Copilot Cal Hacks 12.0, Oct 2025

Claude API, Fish Audio TTS

- Built a Flask backend wiring Claude (Sonnet 4.5) into three endpoints (bedtime-story generation, sleep tips, dream analysis), each with its own prompt template and hard word/token caps to keep responses short enough to read or narrate aloud.
- Wired Fish Audio TTS into the story endpoint to narrate generated stories, handling both response shapes the API can return (hosted URL vs. raw binary audio) and degrading to text-only output if the TTS call fails.

RL Racing Agent In Progress

Python, PPO (stable-baselines3), Pygame/Gymnasium

- Prototyping a reinforcement learning racing agent in a custom Pygame/Gymnasium environment trained with PPO, ahead of porting to BeamNG.tech.